CLASS 8 — SCIENCE · HARD LEVEL UNIT TEST SET 1

Chapters 5–9: **Exploring Forces, Pressure, Winds, Storms and Cyclones, Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures, The Amazing World of Solutes, Solvents and Solutions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Assertion (A): It is easier to push a box on a floor with rollers than on a rough floor. Reason (R): Rolling friction is less than sliding friction. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. Assertion (A): A knife with a sharp edge cuts easily. Reason (R): A sharp edge exerts large pressure because of its small area. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. Assertion (A): Hot tea cools faster when poured in a wide saucer. Reason (R): A larger surface area increases the rate of evaporation. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. Assertion (A): Water is a compound and not a mixture of hydrogen and oxygen. Reason (R): Hydrogen and oxygen in water are chemically combined in a fixed ratio. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. Assertion (A): A solution of sugar in water does not scatter a beam of light. Reason (R): The particles of the solute are of very small size. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. A box of mass 20 kg is pulled with a force of 60 N along a floor where friction is 40 N. The net force is: 1

(a) 20 N **(b)** 100 N **(c)** 60 N **(d)** 40 N

7. A cylinder of weight 120 N stands on a base of radius 10 cm. Taking pi as 3.14, the pressure is about: 1

(a) 38220 Pa **(b)** 12 Pa **(c)** 1200 Pa **(d)** 3822 Pa

8. 100 mL of water and 100 mL of ethanol are mixed. The volume of the mixture is slightly less than 200 mL. This best illustrates: 1

(a) sublimation **(b)** evaporation

(c) condensation **(d)** spaces between particles

9. 4 g of hydrogen combines with 32 g of oxygen to form water. The mass of water formed and the ratio of hydrogen to oxygen are: 1

(a) 36 g, 1 : 4 **(b)** 36 g, 1 : 8 **(c)** 28 g, 1 : 8 **(d)** 36 g, 8 : 1

10. At 40 degrees Celsius, 100 g of water dissolves 64 g of potassium nitrate in a saturated solution. The mass percentage of the saturated solution is about: 1

(a) 100 per cent **(b)** 64 per cent **(c)** 39 per cent **(d)** 36 per cent

11. A tug-of-war rope is stationary. If team A suddenly pulls harder, then the rope: 1

(a) moves towards team B **(b)** moves towards team A

(c) breaks always **(d)** stays at rest

12. Two cyclones, A over warm sea and B moving over land, are compared. Which statement is right? 1

(a) A is likely to be stronger **(b)** B is likely to be stronger

(c) both are equal **(d)** neither has wind

13. **Assertion (A):** Friction is a contact force. 1

Reason (R): Friction acts only when two surfaces are in contact and opposes their relative motion.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

14. **Assertion (A):** A sharp knife cuts better than a blunt one. 1

Reason (R): For the same force, a smaller area of contact gives a larger pressure.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

15. **Assertion (A):** Sugar dissolves in water without much increase in the volume of water. 1

Reason (R): There are spaces between the particles of water into which the sugar particles fit.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. A body of mass 3 kg has a weight of about 30 N on Earth. What is its mass and weight on the Moon if gravity there is one-sixth? 2

17. Calculate the pressure under a 5 kg bag (weight about 50 N) that rests on a 0.1 m by 0.05 m face. 2

18. Explain why a gas does not collect at the bottom of a container although it has weight. 2

19. 4 g of hydrogen reacts completely with oxygen forming 36 g of water. Find the mass of oxygen and the mass ratio of hydrogen to oxygen. 2

20. 100 g of a saturated solution at 60 degrees Celsius contains 37.5 g of solute. Find the mass percentage and the mass of water. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. A box on a floor is pushed with 30 N and moves with a constant velocity. State the friction. If the push is then reduced to 20 N, describe what happens. 3

22. A cube of side 10 cm weighs 50 N. Find the pressure on a table. If a second identical cube is stacked on it, find the new pressure on the table. 3

23. A student heats 50 g of ice from -10 degrees Celsius. Describe, with temperature values, the four stages of the heating curve till 110 degrees Celsius. 3

24. A student has a sample of a grey powder. It is attracted by a magnet and gives hydrogen when acid is added. Identify the powder as an element, compound or mixture and explain. 3

OR

Compare the solubility of NaCl (36 g/100 g water at 20 degrees Celsius) and KNO3 (32 g/100 g water at 20 degrees Celsius). If 100 g water at 20 degrees Celsius is given 40 g of each, find the undissolved mass for each. 3

25. Explain why the friction between a book and the table does not depend much on the area of its face for the same weight. 3

OR

Explain why the speed of wind is greatest near the eye wall. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** Harpreet pulls a box of mass 20 kg across a floor with a horizontal force of 60 N. The force of friction between the box and the floor is 40 N. When he puts rollers under the box, the friction drops to 10 N. 5

(i) Find the net force on the box in the first case.
(ii) Is the box moving or at rest?
(iii) State the direction of friction.
(iv) Find the net force with rollers and explain why rollers help.

27. Design an investigation to find how the temperature affects the rate of diffusion of ink in water. Mention variables, apparatus, observations and conclusion. 5

OR

Design an investigation to distinguish between a mixture and a compound using iron and sulphur. Write the procedure, the observations before and after heating and the conclusion. 5

END OF QUESTION PAPER Total: 50 Marks

MASTERMIND